a Boson: actual metaplectic half-power on A_{group}

columns: $J_2^T g J_2$, J_2^2, and J_1, J_2 commuting (comm.) or anticommuting (anti.)

rows: $J_1^T g J_1$, J_1^2

g, g, comm.	g, g, anti.	g, -g, comm.	g, -g, anti.	-g, g, comm.	-g, g, anti.	-g, -g, comm.	-g, -g, anti.
g, I	B01 d=4 A_{group}: five relations do not fix the half-power cone: none	B02 d=4 A_{group}: five relations do not fix the half-power cone: strict ij_2g^*	B03 d=2 A_{group}: five relations do not fix the half-power cone: none	B04 d=4 anticomm. $(+,-)$: $W_B = \lvert w\rvert^2 \geq 0$ if the pair is Fock-realizable cone: none	B05 d=2 A_{group}: five relations do not fix the half-power cone: strict j_2g^*	B06 d=4 A_{group}: five relations do not fix the half-power cone: none	B07 d=4 anticomm. $(+,-)$: $W_B = \lvert w\rvert^2 \geq 0$ if the pair is Fock-realizable cone: none	B08 d=4 A_{group}: five relations do not fix the half-power cone: none
g, -I	B09 d=2 A_{group}: five relations do not fix the half-power cone: none	B10 d=4 anticomm. $(+,-)$: $W_B = \lvert w\rvert^2 \geq 0$ if the pair is Fock-realizable cone: strict ij_2g^*	B11 d=4 A_{group}: five relations do not fix the half-power cone: none	B12 d=4 anticomm. $(-,-)$: no positive polarization for real orthogonal J_a cone: none	B13 d=4 anticomm. $(+,-)$: $W_B = \lvert w\rvert^2 \geq 0$ if the pair is Fock-realizable cone: strict j_2g^*	B14 d=2 A_{group}: five relations do not fix the half-power cone: none	B15 d=4 anticomm. $(-,-)$: actual $W_B \geq 0$ positive polarization verified at $d=4$ cone: none	B16 d=4 A_{group}: five relations do not fix the half-power cone: none
-g, I	B17 d=2 A_{group}: five relations do not fix the half-power cone: strict ij_2g^*	B18 d=4 A_{group}: five relations do not fix the half-power cone: none	B19 d=4 anticomm. $(+,-)$: $W_B = \lvert w\rvert^2 \geq 0$ if the pair is Fock-realizable cone: none	B20 d=2 A_{group}: five relations do not fix the half-power cone: none	B21 d=4 A_{group}: five relations do not fix the half-power cone: none	B22 d=2 A_{group}: five relations do not fix the half-power cone: strict j_2g^*	B23 d=4 A_{group}: five relations do not fix the half-power cone: none	B24 d=4 anticomm. $(+,-)$: $W_B = \lvert w\rvert^2 \geq 0$ if the pair is Fock-realizable cone: none
-g, -I	B25 d=4 anticomm. $(+,-)$: $W_B = \lvert w\rvert^2 \geq 0$ if the pair is Fock-realizable cone: strict ij_2g^*	B26 d=4 A_{group}: five relations do not fix the half-power cone: none	B27 d=4 anticomm. $(-,-)$: actual $W_B \geq 0$ positive polarization verified at $d=4$ cone: none	B28 d=4 A_{group}: five relations do not fix the half-power cone: none	B29 d=4 A_{group}: five relations do not fix the half-power cone: none	B30 d=4 anticomm. $(+,-)$: $W_B = \lvert w\rvert^2 \geq 0$ if the pair is Fock-realizable cone: strict j_2g^*	B31 d=8 A_{group}: five relations do not fix the half-power cone: none	B32 d=4 anticomm. $(-,-)$: no positive polarization for real orthogonal J_a cone: none

b Fermion: actual Spin half-power on A_{group}

columns: $J_2^T g J_2$, J_2^2, and J_1, J_2 commuting (comm.) or anticommuting (anti.)

rows: $J_1^T g J_1$, J_1^2

g, g, comm.	g, g, anti.	g, -g, comm.	g, -g, anti.	-g, g, comm.	-g, g, anti.	-g, -g, comm.	-g, -g, anti.
g, I	F01 d=2 A_{group}: five relations do not fix the half-power cone: none	F02 d=4 A_{group}: five relations do not fix the half-power cone: none	F03 d=4 A_{group}: five relations do not fix the half-power cone: none	F04 d=4 A_{group}: actual $W_F \geq 0$ anticomm. $(+,-)$: conjugate Majorana blocks, $W_F \geq 0$ cone: strict ij_2g^*	F05 d=2 A_{group}: five relations do not fix the half-power cone: none	F06 d=2 A_{group}: five relations do not fix the half-power cone: none	F07 d=2 A_{group}: actual $W_F \geq 0$ anticomm. $(+,-)$: conjugate Majorana blocks, $W_F \geq 0$ cone: strict j_2g^*	F08 d=2 A_{group}: five relations do not fix the half-power cone: none
g, -I	F09 d=4 A_{group}: five relations do not fix the half-power cone: none	F10 d=4 A_{group}: actual $W_F \geq 0$ anticomm. $(+,-)$: conjugate Majorana blocks, $W_F \geq 0$ cone: none	F11 d=4 A_{group}: five relations do not fix the half-power cone: none	F12 d=8 A_{group}: actual $W_F \geq 0$ anticomm. $(-,-)$: Kramers polarization cone: strict ij_2g^*	F13 d=4 A_{group}: actual $W_F \geq 0$ anticomm. $(+,-)$: conjugate Majorana blocks, $W_F \geq 0$ cone: none	F14 d=4 A_{group}: five relations do not fix the half-power cone: none	F15 d=4 A_{group}: actual $W_F \geq 0$ anticomm. $(-,-)$: Kramers polarization cone: strict j_2g^*	F16 d=4 A_{group}: five relations do not fix the half-power cone: none
-g, I	F17 d=2 A_{group}: five relations do not fix the half-power cone: none	F18 d=2 A_{group}: five relations do not fix the half-power cone: none	F19 d=4 A_{group}: actual $W_F \geq 0$ anticomm. $(+,-)$: conjugate Majorana blocks, $W_F \geq 0$ cone: strict ij_2g^*	F20 d=4 A_{group}: five relations do not fix the half-power cone: none	F21 d=4 A_{group}: five relations do not fix the half-power cone: none	F22 d=4 A_{group}: five relations do not fix the half-power cone: none	F23 d=4 A_{group}: five relations do not fix the half-power cone: none	F24 d=2 A_{group}: actual $W_F \geq 0$ anticomm. $(+,-)$: conjugate Majorana blocks, $W_F \geq 0$ cone: strict j_2g^*
-g, -I	F25 d=2 A_{group}: actual $W_F \geq 0$ anticomm. $(+,-)$: conjugate Majorana blocks, $W_F \geq 0$ cone: none	F26 d=2 A_{group}: five relations do not fix the half-power cone: none	F27 d=4 A_{group}: actual $W_F \geq 0$ anticomm. $(-,-)$: Kramers polarization cone: strict ij_2g^*	F28 d=4 A_{group}: five relations do not fix the half-power cone: none	F29 d=4 A_{group}: five relations do not fix the half-power cone: none	F30 d=2 A_{group}: actual $W_F \geq 0$ anticomm. $(+,-)$: conjugate Majorana blocks, $W_F \geq 0$ cone: none	F31 d=4 A_{group}: five relations do not fix the half-power cone: none	F32 d=4 A_{group}: actual $W_F \geq 0$ anticomm. $(-,-)$: Kramers polarization cone: strict j_2g^*

fermion: conjugate Majorana blocks or Kramers give $W_F \geq 0$ (12/32)

boson: $W_B = \lvert w\rvert^2$ for physical anticomm. $(+,-)$ pairs (8/32)

boson B15/B27: positive polarization verified at d=4

boson B12/B32: impossible for real orthogonal J_a

remaining 20/32: five relations do not fix the actual half-power

Each panel has 8 anticomm. $(+,-)$ classes, 4 anticomm. $(-,-)$ classes, and 20 classes not fixed by the five relations. A_{group} labels assume both symmetries hold for every complete slice. * A strict cone is independent (8/32): nonzero A_{cone} is mapped to $-A_{\text{cone}}$ by Θ_2, so the same configuration is not preserved and the A_{group} proof does not apply.